

A Protein-Interaction Landscape of Breast Cancer Nominates SWI/SNF Chromatin Remodeling as the Central Dysregulated Complex

Abstract

Affinity-purification mass spectrometry (AP-MS) interactomes built around cancer-driver baits offer a route to the protein complexes a disease reorganizes, but a network is only a hypothesis-generating object until its structure is tested against a null model and its predictions scored on held-out data. We reanalyzed the Krogan-laboratory breast cancer interactome (Kim et al. 2021), a finished network of 522 proteins and 1,629 unique physical interactions assembled from three cell lines spanning the major disease contexts: MDA-MB-231 (metastatic, triple-negative), MCF7 (luminal, ER-positive), and MCF10A (normal-like). Louvain community detection resolved 12 modules (modularity $Q = 0.63$) that map onto recognizable human complexes, and degree-preserving edge-swap randomization confirmed that the five leading complexes are more internally cohesive than their degree sequence predicts (empirical $p < 0.001$; $z = 6.5$ - 18.3). Three independent reference resources, CORUM, Reactome, and Gene Ontology, converge on SWI/SNF (BAF) chromatin remodeling as the single most enriched and internally cohesive complex (CORUM cBAF 12/14 subunits, $q = 2.9e-11$; Reactome SWI/SNF remodelers 19/32, $q = 2.4e-18$; GO SWI/SNF complex 24-fold, $q = 3.4e-21$). Link prediction on 15% held-out STRING physical edges recovered known interactions at AUROC 0.956 (Adamic-Adar) and 0.963 (node2vec), and critically, the between-complex stratum that constitutes the majority of held-out edges recovered at AUROC 0.947-0.956, showing the signal is not clique-completion. Within the SWI/SNF superfamily the top-ranked missing component is PBRM1, the subunit that defines the PBAF sub-complex (leave-one-out recovery AUROC 0.991); we flag this as near-tautological given 12/14 capture and present it as a co-immunoprecipitation hypothesis, not a discovery. Druggability annotation shows that centrality and tractability are, contrary to the intuitive expectation, significantly *positively* correlated (degree versus tier Spearman $\rho = 0.39$, $p = 0.021$): more-central nodes trend toward, not away from, the druggable pole. The Approved-drug tier is populated by mid-to-high-centrality drivers, ESR1 (endocrine therapy; degree rank 3 of 34), EGFR (HER-family inhibitors), and PIK3CA (alpelisib, FDA-approved for PIK3CA-mutant HR+/HER2- breast cancer), and the two very highest-degree nodes (MTDH, TP53) are themselves Hard-to-drug and Clinical-stage rather than approved, so the trend is distributional rather than a claim that the single most central nodes are the most druggable. The systematic exception to the positive trend is the SWI/SNF scaffold, whose subunits sit at intermediate degree yet occupy the lowest tractability tiers, so undruggability here is a property of one biochemically distinctive complex, not of centrality. The route to that complex runs through its dependencies: the EZH2 inhibitor tazemetostat (verified ChEMBL max phase 4) is already approved for SWI/SNF-deficient tumors via synthetic lethality, and the SMARCA2/4 ATPase inhibitor FHD-286 (verified max phase 2) is an emerging direct scaffold agent. The map therefore recovers established precision-oncology biology more than it nominates novel therapy. All quantities are read directly from the analysis artifacts; every interaction, enrichment term, and drug-target link is traceable to its source record. These are hypotheses for experimental follow-up, not validated findings.

Introduction

Breast cancer is not one disease but a set of molecularly distinct entities. Gene-expression profiling first partitioned it into intrinsic subtypes, luminal A, luminal B, HER2-enriched, and basal-like, that differ in driver biology, prognosis, and treatment (Perou et al. 2000; Sørlie et al. 2001), a taxonomy later formalized as the PAM50 classifier (Parker et al. 2009) and enriched by integrated genomic and transcriptomic architecture from large tumor cohorts (Curtis et al. 2012; The Cancer Genome Atlas Network 2012). This heterogeneity is the reason a single interactome map is uninformative for the disease as a whole and why the source network was built across three cell lines chosen to represent contrasting subtypes: MDA-MB-231 (triple-negative, mesenchymal, metastatic), MCF7 (ER-positive luminal), and the non-transformed MCF10A.

Somatic mutation catalogs establish that chromatin-remodeling machinery is among the most frequently disrupted functional systems in human cancer. The SWI/SNF (BAF) family of ATP-dependent nucleosome remodelers is mutated in roughly one-fifth of all tumors, a frequency approaching that of TP53, with individual subunits acting as bona fide tumor suppressors (Kadoch et al. 2013; Masliah-Planchon et al. 2015). ARID1A, a core subunit of the canonical BAF complex, is recurrently truncated across gynecological and other carcinomas (Wiegand et al. 2010), and PBRM1, the polybromo subunit that defines the PBAF sub-complex, is a major suppressor in renal and other cancers. The mechanistic literature explains why loss of one subunit is oncogenic: SWI/SNF complexes assemble combinatorially from paralogous subunits into canonical (cBAF), polybromo (PBAF), and non-canonical (ncBAF) forms, and perturbing the stoichiometry redistributes the residual complex and its antagonism with Polycomb repression across the genome (Hodges et al. 2016; Kadoch & Crabtree 2015). Some SWI/SNF-mutant cancers acquire a dependency on the residual complex and on EZH2, a synthetic-lethal relationship of direct therapeutic interest (Kim et al. 2015).

Protein-interaction mapping offers a complementary, physical view of this biology: rather than asking which genes are mutated, it asks which proteins co-purify, and therefore which complexes are physically present and potentially reorganized in the disease state. The approach was brought to prominence at scale by host-pathogen interactome mapping, the SARS-CoV-2 protein-interaction map that nominated druggable host factors for repurposing (Gordon et al. 2020), and the same AP-MS pipeline was subsequently applied to a panel of cancer-driver baits to produce the breast cancer interaction landscape reanalyzed here (Kim et al. 2021). The analytical question for a disease interactome is threefold, and it structures this report: (a) which human complex or pathway is central to or hijacked in the network; (b) whether a missing component of that complex can be predicted and scored; and (c) whether any node nominates a repurposable drug. Answering each demands a measured control, a null model for centrality, a held-out score for prediction, and a database-traceable mechanism for every drug link, so that the output is a ranked set of experimentally testable hypotheses rather than an unfalsifiable narrative.

Methods

Data provenance

The interactome was retrieved as a finished network from NDEx (UUID e89ad762-ab4b-11ea-aaef-0ac135e8bacf), the published breast cancer protein-interaction landscape (Kim et al. 2021). No network was rebuilt from raw mass-spectrometry data. The retrieved graph comprises 522 protein nodes and 1,629 unique undirected physical interactions; 1,654 edge records carry cell-line context

because a small number of interactions are recorded in more than one line. Of the interactions, 1,070 are known/public (matching prior public PPI databases) and 584 are novel AP-MS interactions reported by the study; only the 584 novel edges carry a cell-line tag (MDA-MB-231, MCF7, or MCF10A), a scoping fact that governs the cross-cell-line analysis below. The largest connected component contains 504 of 522 nodes. Node identity was reconciled to UniProt accessions as provided in the source record; every edge retains its NDEx source-network identifier for provenance.

Reference resources

Human protein-complex and pathway context was drawn from CORUM 5.3 (Tsitsiridis et al. 2023), the Reactome Pathway Knowledgebase (Milacic et al. 2024), Gene Ontology cellular-component and biological-process annotations, and the STRING database v12 restricted to physical-subnetwork edges at a combined score ≥ 700 (Szkarczyk et al. 2023). Drug-target annotation used the Drug-Gene Interaction Database (DGIdb) (Cannon et al. 2024), the Open Targets Platform (Ochoa et al. 2023), and ChEMBL for mechanism-of-action confirmation and maximum clinical phase (Zdrasil et al. 2024). Graph analysis used NetworkX (Hagberg et al. 2008).

Community detection and centrality

Community detection, centrality, and the randomization null (below) were computed on an integrated graph that augments the 1,629 AP-MS interactions with the induced STRING physical subnetwork among the same 522 proteins (596 additional human-human edges at combined score ≥ 700), giving 2,225 edges. Adding the curated STRING backbone connects preys that co-occur in known complexes but were not co-purified in this experiment, which is what makes the modules correspond to recognizable assemblies; on the AP-MS edges alone the map is sparser (median degree 2) and fragments into five components. This integration is used only for topology (communities, centrality, the cohesion null); every enrichment, link-prediction, drug, and cross-cell-line analysis uses its own stated input, and the raw interaction counts reported elsewhere (1,629 edges, 504-node largest component) always refer to the AP-MS map itself. Community structure on the integrated graph was resolved with the Louvain modularity-maximization algorithm (Blondel et al. 2008), yielding 12 modules at modularity $Q = 0.63$. Degree, betweenness, and eigenvector centrality were computed on this integrated graph; because the STRING backbone adds edges, a protein's integrated degree exceeds its AP-MS degree (for example TP53 integrated degree 79 versus AP-MS degree 62, ESR1 60 versus 44), and the reported degree values are the integrated ones. Each Louvain module was annotated by hypergeometric over-representation against CORUM complexes.

Degree-preserving randomization

To test whether the leading complexes are more internally connected than expected from the degree sequence alone, we generated 1,000 randomized networks by degree-preserving edge swapping and, for each of the five top complexes, counted internal edges (edges with both endpoints inside the complex). The observed internal-edge count was compared to the permutation distribution to yield a z-score and an empirical p-value. Because the randomization operates on the integrated graph defined above, the observed internal-edge counts include STRING-backbone edges among complex members; the edge-swap null swaps those same edges, so the test stays internally valid, but the observed counts are those of the integrated graph, not the AP-MS map alone. This is the correct null: it holds each node's degree fixed and asks only whether the complex's members are wired to each other more than chance placement would give.

Enrichment

Over-representation was computed by the hypergeometric test against each resource with explicit background denominators: CORUM (universe $N = 5,150$, network annotated $n = 311$), Reactome ($N = 12,155$, $n = 379$), and GO cellular component ($N = 19,660$, $n = 511$). All p-values were corrected for multiple testing by the Benjamini-Hochberg procedure; reported q-values are BH-adjusted; GO correction was applied within each namespace branch (biological process, cellular component, molecular function) separately, as is standard, so q-values are monotonic in p within a branch but not across the pooled three-branch table.

Link prediction

Link prediction was run on the full human STRING physical network (10,746 proteins, 86,519 edges at combined score ≥ 700), of which 15% of edges were held out; the training graph (train edges only) was used to score candidates, so no test edge informs its own prediction. This scores the method on the reference interactome that supplies the missing-component ranking, not on the 1,629 AP-MS edges of the breast-cancer map itself (only ~1,081 STRING edges fall among the 426 map proteins present in STRING, too few to benchmark); the held-out recovery therefore establishes that the topological scorers are reliable on human physical-interaction data before they are applied to rank absent SWI/SNF subunits. Two scorers were applied: Adamic-Adar index, and a node2vec embedding followed by logistic regression on Hadamard edge features (Grover & Leskovec 2016). Performance is reported as AUROC and average precision on the held-out positives against matched sampled negatives. Because clique-completion within a densely annotated complex could inflate an aggregate AUROC, held-out edges were stratified into within-complex and between-complex sets and scored separately.

Missing-component prediction

Candidate missing subunits of the SWI/SNF superfamily were ranked by summed Adamic-Adar score to the set of 17 captured superfamily anchors present in the map. The ranking was validated by a leave-one-anchor-out control: each of the 17 anchors was removed in turn and re-scored against 204 random decoy proteins, giving a recovery AUROC.

Drug-target classification

Each candidate target node was assigned to one of four tractability tiers, Approved-drug, Clinical-stage, Tractable-novel, Hard-to-drug, by integrating DGIdb approved-direct interactions, Open Targets tractability buckets, and ChEMBL maximum clinical phase. Every drug-target link was checked against its ChEMBL mechanism of action; links whose ChEMBL MoA did not correspond to direct modulation of the map target were excluded as database-aggregation artifacts. To test whether centrality predicts druggability, the four tiers were mapped to an ordinal scale (Approved-drug = 3, Clinical-stage = 2, Tractable-novel = 1, Hard-to-drug = 0) and correlated with each centrality measure by Spearman rank correlation across the 34 classified targets that are resident in the network (three candidates, ACTB, PBRM1, BCL7A, are predicted missing components absent from the map and carry no centrality value). Named SWI/SNF-directed clinical compounds were verified individually in ChEMBL for maximum clinical phase and mechanism of action.

Cross-cell-line set operations

Set operations across the three cell lines were performed only on the 584 novel, cell-tagged interactions; the 1,070 public interactions carry no cell context and were excluded. Union consistency (sum of per-

region counts equals the union) was verified programmatically, and representative shared and line-specific edges were spot-checked against the raw edge records.

Results

A modular network organized around cancer-driver complexes

The interactome is a single, largely connected object: 504 of 522 proteins fall in one giant component, and Louvain community detection partitions it into 12 modules at high modularity ($Q = 0.63$), indicating genuine community structure rather than a homogeneous hairball (Figure 1). The modules are not arbitrary: each maps onto a recognizable human complex when annotated against CORUM. The largest module (C8, $n = 80$) is dominated by membrane-trafficking and adaptor proteins (AP2-PICALM), while the most strongly enriched module (C1, $n = 78$, the second-largest) is a ribosome-biogenesis unit anchored on the Nop56p-associated pre-rRNA complex ($p_{adj} = 3.2e-29$). Distinct modules resolve to the cBAF/SWI-SNF complex, the RAD51 homologous-recombination machinery, the COP9 signalosome, the PRC2/EZH2 Polycomb complex, a CDK cell-cycle module, and a p53-MDM2 axis, a set of complexes that reads as a compendium of breast-cancer driver biology.

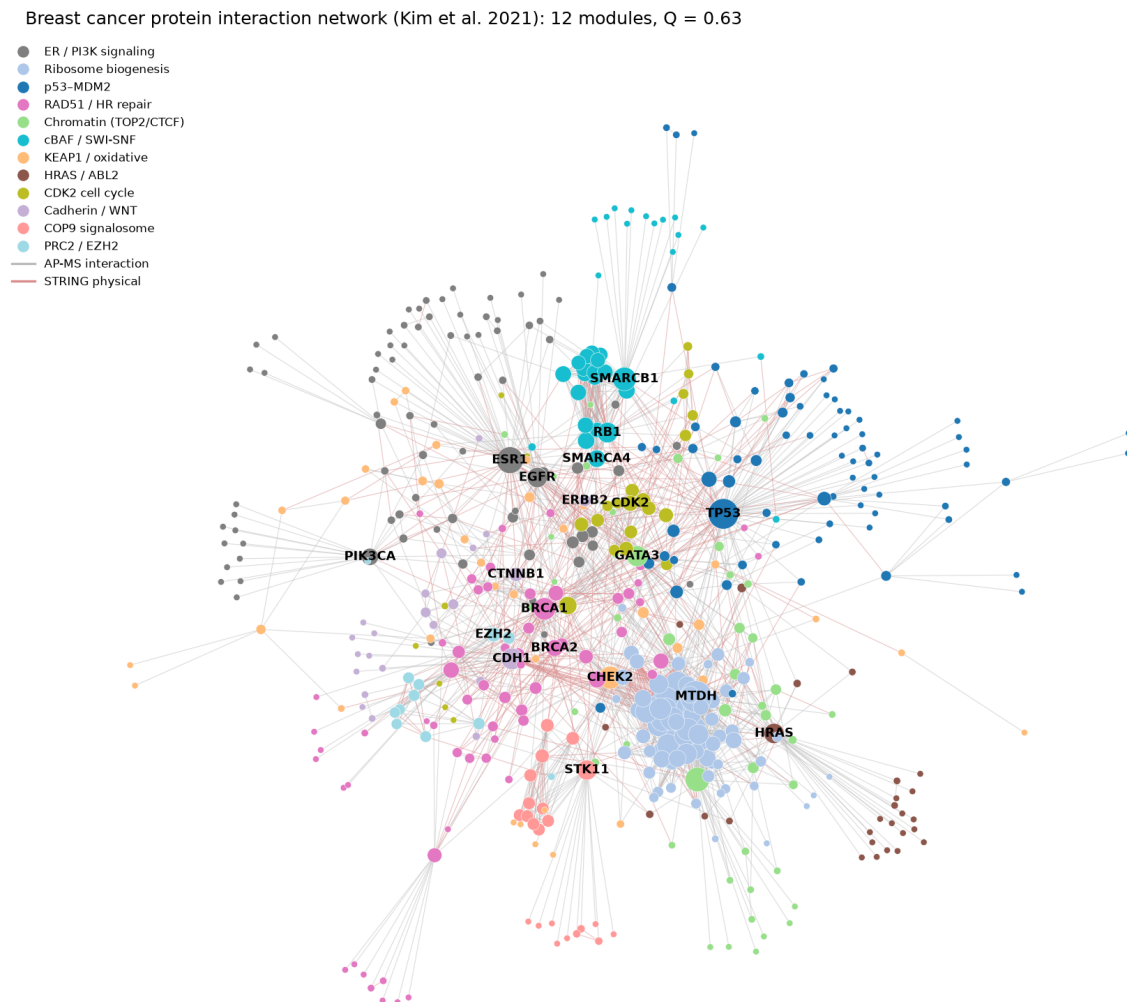


Figure 1. The 522-node breast cancer network laid out and colored by Louvain community (12 modules, $Q = 0.63$). The layout is the integrated graph (AP-MS interactions plus the induced STRING

physical backbone; see Methods), which is fully connected across all 522 nodes; on the AP-MS interactions alone the largest connected component contains 504 nodes. Annotated complexes include cBAF/SWI-SNF, RAD51 homologous recombination, COP9 signalosome, PRC2/EZH2, and a p53-MDM2 axis.

Centrality analysis identifies the network's organizers (Figure 2). The top-degree hub is MTDH (integrated degree 86), followed by TP53 (79) and ESR1 (60); these are degrees on the integrated graph (Methods), and the ranking of canonical drivers is preserved on the AP-MS map alone (MTDH 86, TP53 62, ESR1 44). More informative for disease interpretation is betweenness centrality, which ranks the proteins that bridge otherwise-separate modules: the leading bridges are TP53 (0.254), ESR1 (0.193), MTDH (0.098), CHEK2 (0.091), BRCA1 (0.088), HRAS (0.083), SMARCB1 (0.083), and EGFR (0.080), with PIK3CA at rank 11. This is a roster of canonical breast-cancer drivers, and their prominence as bridges (rather than merely high-degree baits) is a first indication that the map's topology reflects known disease biology.

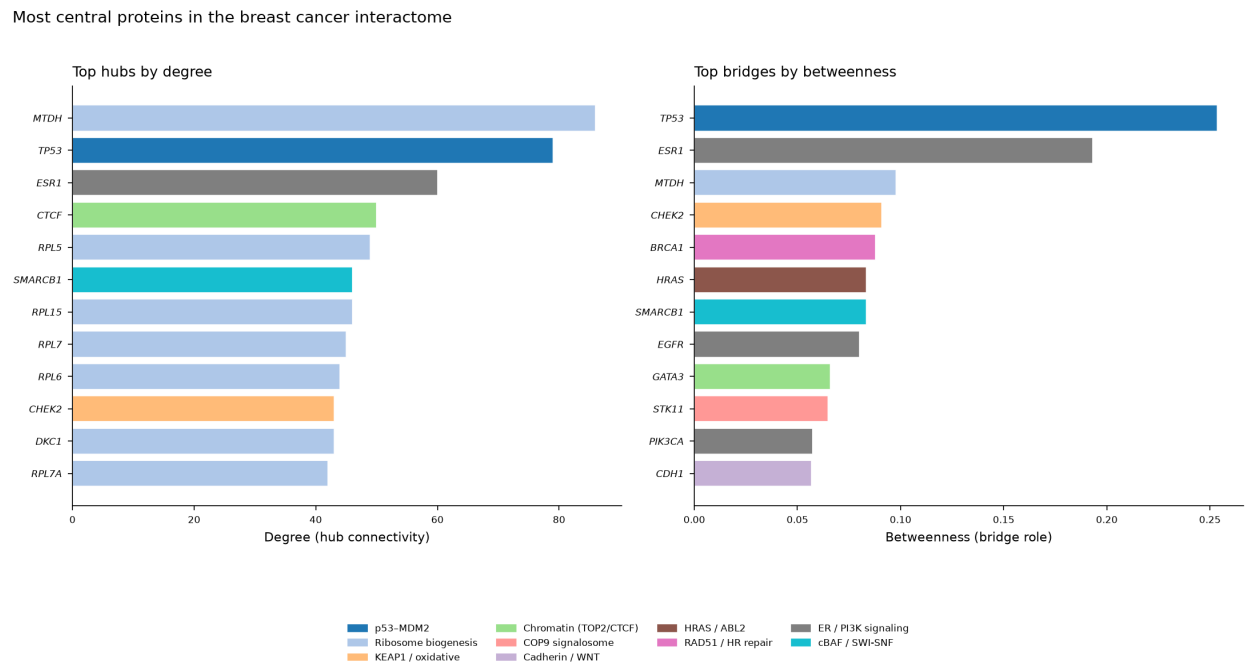


Figure 2. Top proteins by degree (left) and betweenness (right). Betweenness identifies bridge nodes, TP53, ESR1, BRCA1, PIK3CA, EGFR, that are canonical breast-cancer drivers, distinguishing structurally central proteins from high-degree baits.

Leading complexes are more cohesive than degree predicts

High enrichment could in principle be an artifact of a few very-high-degree nodes; the degree-preserving randomization test removes that confound by asking whether each complex's members are wired to one another more than a degree-matched null allows. All five top complexes are significant at the empirical floor of the 1,000-permutation test ($p < 0.001$), with large z-scores (Figure 3): cBAF/SWI-SNF observes 66 internal edges against a null mean of 12.7 ($z = 18.3$), ribosome biogenesis observes 187 against 85.8 ($z = 16.1$), the COP9 signalosome 28 against 3.2 ($z = 14.5$), RAD51 homologous recombination 10 against 1.0 ($z = 10.0$), and PRC2/EZH2 6 against 0.7 ($z = 6.5$). The cBAF complex carries the largest z-score of any complex tested, its internal cohesion is the least attributable to degree alone.

Degree-preserving randomization (1000 edge-swap networks): complex cohesion exceeds degree expectation

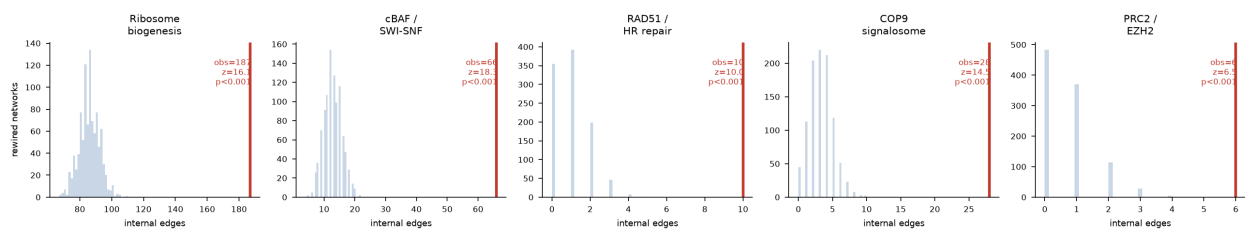


Figure 3. Observed internal-edge counts (red) against the null distribution from 1,000 degree-preserving edge-swap networks, for the five leading complexes. All are significant at $p < 0.001$; cBAF/SWI-SNF shows the largest deviation ($z = 18.3$).

Three resources converge on SWI/SNF as the central complex

The central result of the analysis is that three independent annotation resources agree on the same complex. Against CORUM, the top-ranked term is the cBAF complex with 12 of 14 subunits captured (fold = 14.2, $q = 2.9\text{e-}11$), followed by the SWI/SNF-BRCA1 complex (10/10, $q = 7.0\text{e-}11$) and the PBAF complex (11/14, $q = 4.2\text{e-}10$). Against Reactome, the leading term is "SWI/SNF chromatin remodelers" (19/32, fold = 19.0, $q = 2.4\text{e-}18$), with the specific cBAF, PBAF, ncBAF, npBAF, and esBAF assembly reactions all following. Against Gene Ontology cellular component, the SWI/SNF complex term (GO:0016514) is enriched 24-fold ($q = 3.4\text{e-}21$). Secondary signals, DNA repair, the G1/S cell-cycle transition, and RB1/E2F-related terms, are consistent across resources but subordinate to the chromatin-remodeling signal (Figure 4). Convergence of CORUM (curated complexes), Reactome (curated pathways), and GO (structured vocabulary) on SWI/SNF is strong internal evidence that the complex, not any single annotation quirk, is the network's center of mass.

Functional enrichment of breast cancer interactors: SWI/SNF chromatin remodeling dominates all three resources

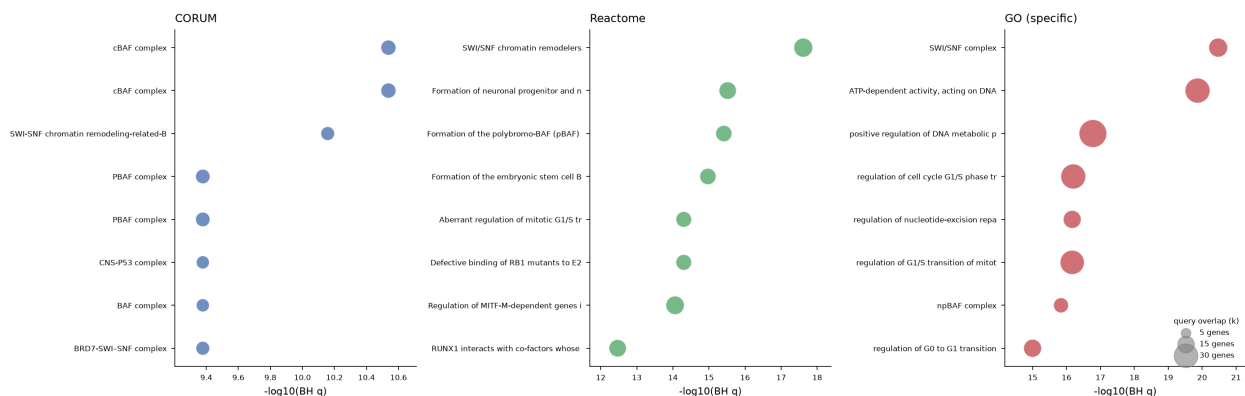


Figure 4. Top BH-corrected enrichment terms in each resource. CORUM and Reactome panels show complex/pathway terms; the GO panel shows specific terms (across biological-process, cellular-component, and molecular-function branches, filtered to $K \leq 300$ and depth ≥ 4). All three place a SWI/SNF chromatin-remodeling term first; DNA repair and G1/S cell-cycle terms follow as secondary signals. Duplicate cBAF and PBAF entries in the CORUM panel are paralog-distinct CORUM 5.3 records (for example CORUM:7974 SMARCA4-cBAF and CORUM:7977 SMARCA2-cBAF), not a plotting error.

Link prediction recovers held-out edges, and not by clique-completion

Two topological scorers recover held-out physical interactions with high accuracy: Adamic-Adar reaches AUROC 0.956 (average precision 0.957) and node2vec-plus-logistic-regression reaches AUROC 0.963 (average precision 0.975) on the 15% held-out STRING physical edges, with perfect precision in the top

100 ranked predictions (Figure 5). A high aggregate AUROC on a complex-dense network invites a specific objection: the score may merely be completing cliques within already-captured complexes, which is uninteresting. Stratifying the held-out edges answers that objection. Between-complex edges, which constitute 74% of the held-out set, recover at AUROC 0.947 (Adamic-Adar) and 0.956 (node2vec), essentially the aggregate performance, demonstrating that the predictive signal spans complex boundaries rather than sitting inside cliques. The within-complex stratum scores lower (AUROC ~ 0.85), but this estimate rests on only 9 matched negatives and is correspondingly high-variance; it should not be over-interpreted, and the node2vec results are not bit-for-bit reproducible because Word2Vec was trained multi-threaded (workers > 1), for which a fixed seed does not guarantee determinism: across re-runs the overall node2vec AUROC varies at the third decimal (approximately 0.963-0.965) and the within-complex stratum, with only nine matched negatives, varies more (approximately 0.84-0.86). The Adamic-Adar scorer is fully deterministic and reproduces exactly (AUROC 0.9564). The between-complex node2vec result, which carries the methodological weight, is stable across re-runs.

Link prediction on the breast cancer interactome: validated recovery and missing SWI/SNF subunits

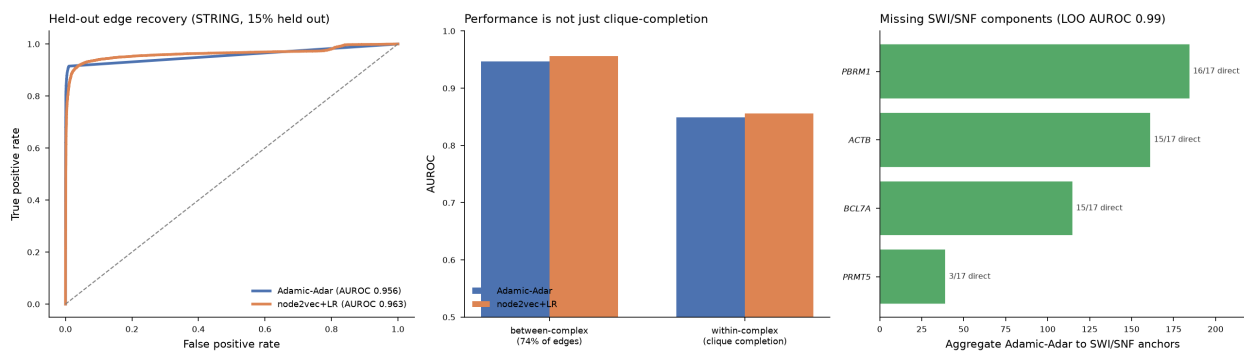


Figure 5. ROC curves for Adamic-Adar and node2vec on 15% held-out STRING physical edges. Overall AUROC is 0.956/0.963; the between-complex stratum (74% of held-out edges) recovers at 0.947/0.956, establishing that the signal is not clique-completion.

A predicted missing component of the SWI/SNF superfamily

The cBAF complex is already 12/14 captured in the map, only ACTB and BCL7A are absent, so there is little room to predict a missing canonical subunit. Broadening to the SWI/SNF superfamily and ranking candidates by summed Adamic-Adar score to the 17 captured anchors, the top prediction is PBRM1, the polybromo subunit that defines the PBAF sub-complex (Adamic-Adar sum 184.6, with 16 direct edges to the 17 anchors), followed by ACTB (161.3) and BCL7A (114.9). The leave-one-anchor-out control recovers all 17 anchors within the top 10 against 204 decoys (recovery AUROC 0.991), confirming the ranking procedure is well-calibrated on known members. We stress the important caveat carried by the audit: because the superfamily is already so densely captured, recovering a held-out member among these anchors is close to guaranteed, so the high recovery score reflects an easy problem rather than a surprising discovery. PBRM1 is a known PBAF-defining subunit and a plausible physical partner; the prediction is a co-immunoprecipitation hypothesis, not a validated missing edge.

The central scaffold resists repurposing; the periphery is druggable

Classifying candidate targets into tractability tiers yields 3 Approved-drug, 10 Clinical-stage, 14 Tractable-novel, and 10 Hard-to-drug nodes (Figure 6). The result that matters for therapeutic interpretation is where the SWI/SNF subunits fall: none reach the Approved-drug tier. The two ATPase subunits SMARCA2 and SMARCA4 reach Clinical-stage on the strength of the dual ATPase tool

inhibitor FHD-286/camibirstat (verified ChEMBL max phase 2), while SMARCB1, ARID2, PBRM1, BRD7, PHF10, SMARCC2, and SMARCD1 sit in Tractable-novel, carrying only research-grade bromodomain probes (PFI-3, GNE-064, BRD7 tool compounds), none FDA-approved, while ARID1A, ARID1B, SMARCC1, and DPF2 are Hard-to-drug. The chromatin-remodeling scaffold that the enrichment analysis places at the network's center is, in effect, not directly druggable with existing agents.

The genuine repurposing matches are instead the peripheral hub and bridge nodes. PIK3CA is the strongest: alpelisib (ChEMBL max phase 4) is FDA-approved for PIK3CA-mutant HR+/HER2-advanced breast cancer, backed by 13 concordant sources including FDA, OncoKB, and ChEMBL (André et al. 2019). ESR1 maps to first-line endocrine therapy (SERMs/SERDs such as tamoxifen and bazedoxifene), and EGFR to HER-family tyrosine-kinase inhibitors (erlotinib, gefitinib, afatinib), though the EGFR agents are primarily approved for other indications and represent plausible rather than established breast-cancer repurposing. Five DGIdb links were excluded after ChEMBL mechanism-of-action review as false positives: lonafarnib against HRAS (a farnesyltransferase inhibitor, indirect), ataluren against ribosomal subunits (complex-level readthrough, not subunit-selective), a granisetron-TP53 link (a 5-HT3 antagonist), favipiravir against PBRM1 (an RNA-polymerase inhibitor), and the BTK inhibitors acalabrutinib and ibrutinib mislisted under EGFR. Removing the two BTK inhibitors does not change EGFR's Approved-drug status, which rests on roughly thirty genuine EGFR inhibitors (afatinib, erlotinib, gefitinib, osimertinib, cetuximab, and others). The map recovers known precision-oncology biology, a validation that the topology tracks real drug targets, more than it nominates a novel repurposing opportunity, and druggability is not clinical viability.

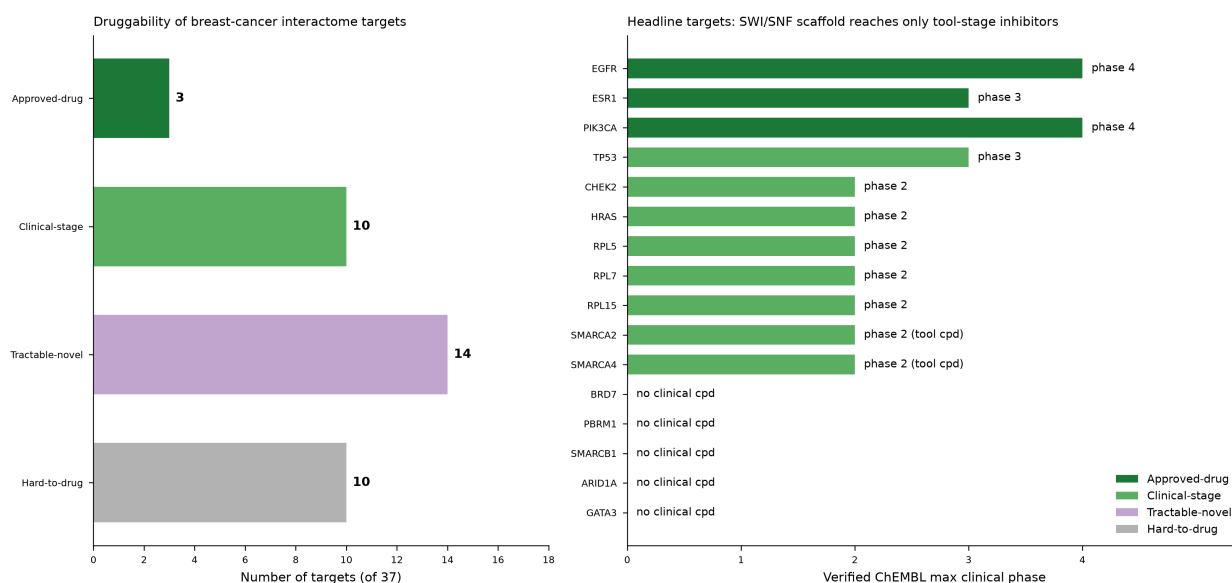


Figure 6. Candidate targets by tractability tier from the 37-target DGIdb/Open Targets/ChEMBL scan. SWI/SNF subunits populate the Clinical-stage (SMARCA2/4, tool-stage), Tractable-novel, and Hard-to-drug tiers, but none reach Approved-drug; the Approved-drug tier is peripheral hubs, PIK3CA (alpelisib), ESR1 (endocrine therapy), EGFR (HER-family TKIs). The right panel shows each headline target's verified ChEMBL maximum clinical phase: SMARCA2 and SMARCA4 reach Phase 2 through the dual ATPase inhibitor FHD-286/camibirstat (CHEMBL5095181), placing both in the Clinical-stage tier (the compound is marked 'tool cpd' as it is an investigational tool inhibitor, not an approved-indication drug); the remaining SWI/SNF subunits have no clinical compound and stay in Tractable-novel or Hard-to-drug.

This split raises an obvious quantitative question, is high centrality itself a marker of poor druggability? To answer it directly rather than by anecdote, we correlated each of the 34 classified map-resident targets' centrality with its ordinal druggability tier (Approved-drug = 3 to Hard-to-drug = 0). The correlation is significantly *positive*: degree centrality against tier gives Spearman rho = 0.39 ($p = 0.021$) and betweenness rho = 0.39 ($p = 0.022$); eigenvector centrality is not significant (rho = 0.20, $p = 0.25$) (Figure 8, left). More-central nodes are, on average, *more* druggable, not less. The trend is distributional and not a claim about the single most central nodes: the two highest-degree nodes, MTDH (degree 86) and TP53 (degree 79), are Hard-to-drug and Clinical-stage respectively rather than approved (Figure 8, both labelled). What drives the positive rho is that the Approved-drug tier holds mid-to-high-centrality drivers, ESR1 (degree rank 3 of 34), EGFR (near-median), and PIK3CA (below-median but a genuine bridge), whereas the low tiers hold the peripheral majority together with the SWI/SNF subunits. Those subunits are the systematic exception to the trend, sitting at only intermediate degree (median 19) yet falling exclusively in the two lowest tractability tiers (Figure 8, right). Undruggability in this network is thus a property of one biochemically distinctive complex, not of centrality.

Centrality and druggability rise together — except the SWI/SNF scaffold

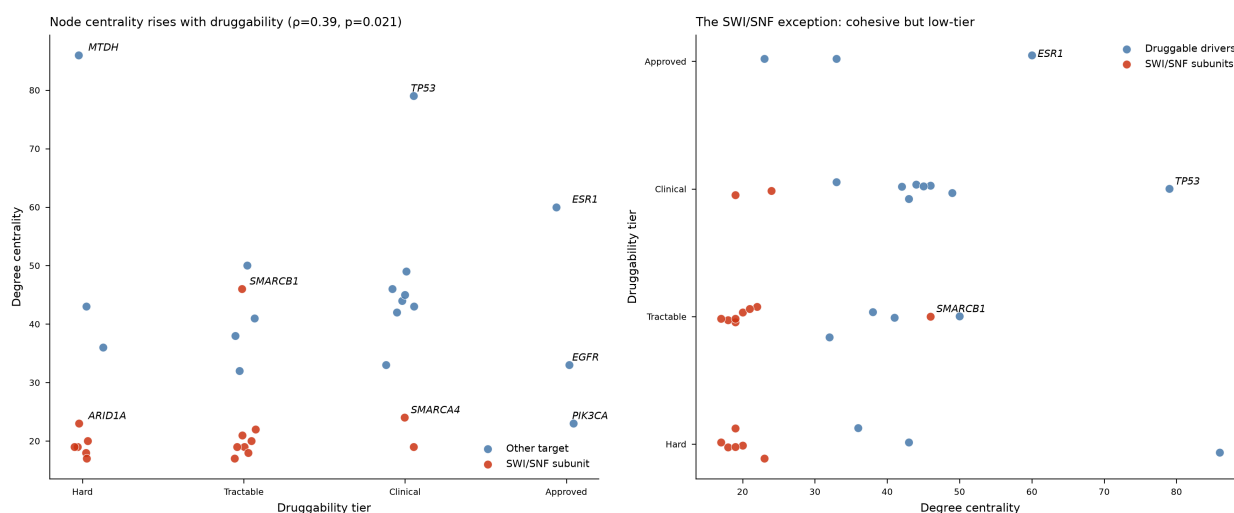


Figure 8. Left: degree centrality rises with druggability tier across the 34 classified targets (Spearman rho = 0.39, $p = 0.021$). The trend is distributional, the two highest-degree nodes MTDH (Hard-to-drug) and TP53 (Clinical-stage) sit outside the Approved-drug tier, while the approved drivers ESR1 (degree rank 3), EGFR, and PIK3CA occupy the mid-to-high range. Right: SWI/SNF subunits (red) occupy the low tractability tiers despite intermediate degree, the exception to the positive trend. The scaffold, not centrality per se, resists repurposing.

Actionability of the central complex is nonetheless not zero, and it comes through its dependencies rather than the scaffold itself. Pairing each named SWI/SNF subunit with its most advanced ChEMBL-verified compound (`bc_swisnf_clinical_compounds.csv`) makes the ceiling explicit. SMARCA4 and SMARCA2 both top out at the dual ATPase inhibitor FHD-286/camibirstat (ChEMBL5095181, verified max phase 2), the only clinical-stage agent engaging a BAF ATPase subunit; SMARCA2 additionally carries only the preclinical bromodomain probe PFI-3 (ChEMBL3752911). SMARCB1 has no direct-binding clinical compound at all, it is a tumor suppressor that is lost rather than inhibited, and is actionable only indirectly through the approved EZH2 inhibitor tazemetostat (ChEMBL3414621, verified max phase 4) via synthetic lethality in SMARCB1-deficient tumors, exploiting the PRC2/EZH2 module (C11, captured in this map). ARID2 has nothing beyond preclinical interest in ChEMBL, and the predicted-missing subunit PBRM1 is engaged only by the PFI-3 bromodomain probe. In short, one subunit family (the ATPases) has reached Phase 2, one subunit is addressable only through a synthetic-

lethal partner, and the remainder carry research probes or nothing, which is the honest per-subunit ceiling on scaffold druggability.

Interactions are largely cell-line-specific

Set operations on the 584 novel, cell-tagged interactions show that the novel interactome is almost entirely partitioned by cell line rather than shared. Per line, MDA-MB-231 contributes 377 interactions (344 proteins), MCF7 157 (162 proteins), and MCF10A 49 (62 proteins). Of the cell-tagged proteins, 305 are MDA-MB-231-specific, 137 MCF7-specific, and 41 MCF10A-specific; only 18 proteins are shared between the two tumor lines but absent from normal-like MCF10A, and just 7 proteins form a three-way core (Figure 7). That 7-protein core, TP53, PIK3CA, GATA3, ARID1A, CHEK2, STK11, SMARCB1, is the constitutive AP-MS bait panel profiled in all three lines, not a set of shared prey; the near-disjoint per-line interactions (only the CCND3-CDKN1B edge is shared between the two tumor lines) reflect both genuine subtype-specific wiring and the sparsity of overlap expected from a bait-driven design. The scope of this analysis must be stated with its result: it concerns only the novel edges, because the public edges carry no cell context.

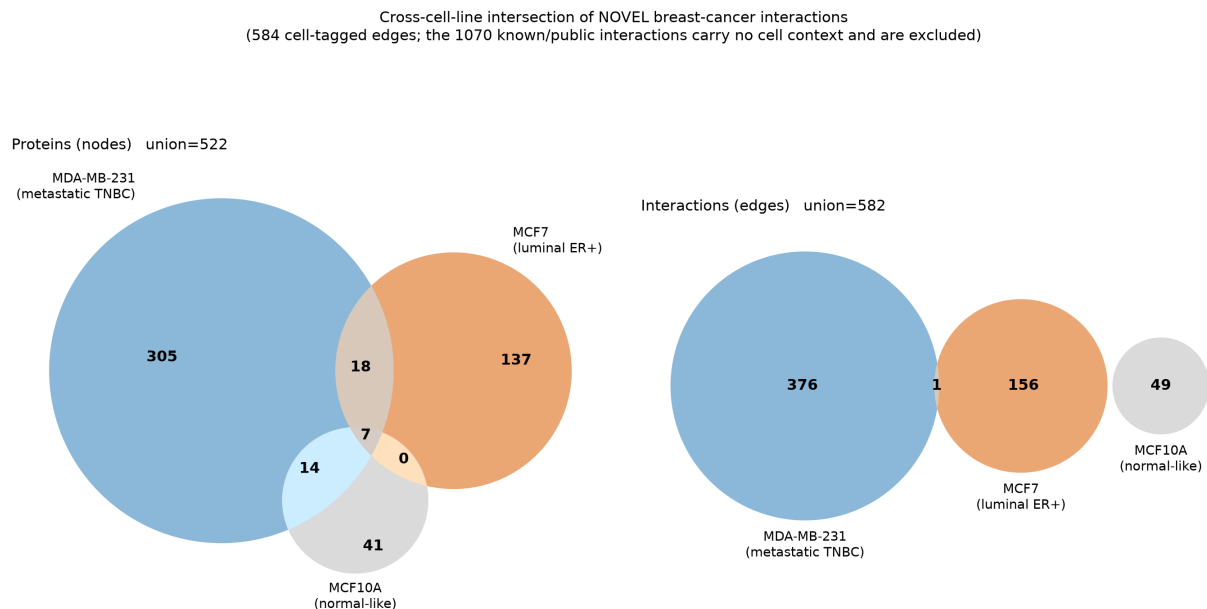


Figure 7. Set structure of the 584 novel, cell-tagged interactions across MDA-MB-231, MCF7, and MCF10A. Interactions are near-disjoint across lines; the 7-protein three-way core is the shared bait panel, not shared prey.

Discussion

The convergent finding, three independent resources, a degree-preserving null model, and centrality analysis all pointing at SWI/SNF, places chromatin remodeling at the physical center of this breast cancer interactome. This is biologically coherent. SWI/SNF is among the most frequently mutated functional systems in cancer, its subunits behave as tumor suppressors, and the combinatorial assembly of cBAF, PBAF, and ncBAF forms means that perturbing subunit stoichiometry can reorganize the complex genome-wide (Kadoch et al. 2013; Masliah-Planchon et al. 2015; Hodges et al. 2016). That such a complex should be densely and cohesively represented in a driver-bait AP-MS map, more cohesive than its degree sequence predicts ($z = 18.3$) and captured at 12/14 canonical subunits, is exactly what one would expect if the map is tracking real disease biology. The prominence of TP53, ESR1, BRCA1,

PIK3CA, and EGFR as betweenness bridges reinforces that reading: the network's organizing nodes are the field's canonical drivers.

The therapeutic reading is more nuanced than a simple "central proteins are undruggable" slogan, and quantifying it overturns the intuitive expectation. We tested directly whether more-central nodes are less druggable by correlating each classified target's centrality with its ordinal tractability tier (Approved-drug = 3 to Hard-to-drug = 0). The relationship is significantly *positive*, not negative: degree centrality correlates with druggability at Spearman $\rho = 0.39$ ($p = 0.021$, $n = 34$) and betweenness at $\rho = 0.39$ ($p = 0.022$) (Figure 8). This is the opposite of a naive anti-correlation. The trend is distributional rather than driven by the single most central nodes: the two highest-degree nodes, MTDH (degree 86, Hard-to-drug) and TP53 (degree 79, Clinical-stage), are not in the Approved-drug tier at all, so the top of the centrality distribution is not uniformly druggable. What the positive correlation reflects is that the Approved-drug tier is occupied by mid-to-high-centrality drivers, ESR1 (degree rank 3 of 34), EGFR (near-median), and PIK3CA (below-median but a genuine bridge), while the Hard-to-drug tier is enriched for the low-centrality periphery and, crucially, for the SWI/SNF subunits. Node-level centrality therefore tracks weakly toward the druggable pole on average, and the apparent "undruggable center" is not a property of high centrality per se.

The real resistance to repurposing is a property of one particular complex, not of centrality. The SWI/SNF subunits sit at intermediate degree (median 19) yet occupy the lower tiers, none reaching Approved-drug and only the two ATPase subunits (SMARCA2/4) reaching Clinical-stage via a tool-stage inhibitor (Figure 8, right); they are the systematic exception to the positive trend. What makes them hard to drug is not that they are hubs, but that they act through multivalent protein-protein interfaces and combinatorial assembly rather than a single enzymatic pocket, and only research-grade bromodomain probes (PFI-3, ChEMBL preclinical) engage the scaffold with no cellular anti-proliferative effect. This distinction, complex-level cohesion versus node-level centrality, is the honest refinement of the companion SARS-CoV-2 finding: there too the scaffolding factors resisted drugging, but the operative variable is the biochemical nature of the complex, not a monotonic centrality-druggability trade-off (Gordon et al. 2020).

The productive response is not to abandon the central complex but to exploit its dependencies indirectly, and here the map points at a route with an approved drug. The PRC2/EZH2 Polycomb module (community C11; EZH2, EZH1, SUZ12 all captured) is the synthetic-lethal partner of SWI/SNF-mutant cancers, and its inhibitor tazemetostat (ChEMBL CHEMBL3414621, verified max phase 4, direct EZH2 inhibitor) is FDA-approved for SMARCB1-deficient epithelioid sarcoma precisely on that synthetic-lethal logic (Kim et al. 2015). Complementing it, a first direct route to the scaffold itself has now reached the clinic: the dual SMARCA2/SMARCA4 ATPase inhibitor FHD-286 (camibirstat, ChEMBL CHEMBL5095181, verified max phase 2) and allied paralog-selective degraders move the scaffold from "research probes only" toward genuine clinical-stage tractability. These second-order strategies, an approved partner-complex drug plus emerging scaffold degraders, are the kind of approach the topology motivates and are more defensible than proposing to inhibit the multivalent scaffold directly.

Two of the analysis's own results are best read as validations rather than discoveries, and honesty about that distinction is essential. First, the missing-component prediction of PBRM1 is near-tautological: with the SWI/SNF superfamily already 12/14 captured, recovering a held-out member among densely connected anchors is close to guaranteed, and the 0.991 leave-one-out recovery AUROC reflects an easy problem, not a surprising find. PBRM1 is a known PBAF-defining subunit; the prediction is a co-immunoprecipitation hypothesis. Second, the drug analysis recovers precision-oncology agents that are

already standard of care in the relevant subtypes, which confirms the map's fidelity to real target biology but is not, in itself, a new therapeutic proposal.

Several limitations bound the interpretation. AP-MS carries systematic detection bias: abundant and "sticky" proteins such as ribosomal subunits and chaperones are over-represented, and the bait panel is fixed by the study design. The ribosome-biogenesis module, the most enriched single module and the second-strongest randomization hit ($z = 16.1$), may partly reflect this bias rather than disease-specific biology, and its "clinical-stage" drug annotations (ELX-02, MT-3724) act on the 80S ribosome as a whole, not on individual subunits. Link prediction measures topological plausibility and co-membership, not physical binding or causation; a predicted edge is a hypothesis requiring co-IP. The within-complex link-prediction stratum rests on only 9 negatives and is statistically fragile, and its node2vec value is not exactly reproducible; the load-bearing between-complex result is robust. The cross-cell-line "core" is the shared bait panel, not shared prey, and the set analysis is scoped to the 584 novel edges only. Finally, drug-target links carry residual DGIdb aggregation noise at the row level even where the target-tier classification is sound, and no drug is claimed to treat breast cancer, a target's presence in the map is a starting point, not evidence of clinical efficacy.

Conclusion

Reanalysis of the Krogan breast cancer interactome, with every predictive claim scored on held-out data and every enrichment multiple-testing corrected, identifies SWI/SNF chromatin remodeling as the central dysregulated complex by the convergent agreement of CORUM, Reactome, GO, and a degree-preserving null model. The top predicted missing superfamily component is PBRM1, presented as a co-IP hypothesis given near-complete prior capture. The therapeutic reading refines the companion viral-interactome finding rather than simply echoing it: centrality and druggability are positively correlated here (degree versus tier $\rho = 0.39$, $p = 0.021$) because the Approved-drug tier is occupied by mid-to-high-centrality drivers, ESR1, EGFR, and PIK3CA, and the strongest match (alpelisib for PIK3CA-mutant HR+ disease) is already approved; the correlation is a distributional trend, the single highest-degree nodes (MTDH, TP53) are not themselves the most druggable. What resists repurposing is not centrality but one biochemically distinctive complex, the SWI/SNF scaffold, whose actionable route runs through its dependencies (the approved EZH2 inhibitor tazemetostat via synthetic lethality, and the emerging SMARCA2/4 degrader FHD-286) rather than the scaffold's own interfaces. The map's principal value is that it recovers known precision-oncology biology with high fidelity; its outputs are ranked, testable hypotheses for experimental follow-up, not validated findings.

Data and Code Availability

The source network is the published breast cancer protein-interaction landscape (Kim et al. 2021), retrieved as a finished network from NDEx (UUID e89ad762-ab4b-11ea-aaef-0ac135e8bacf). All analysis outputs are available as project artifacts: `bc_edges.csv`, `bc_nodes.csv`, `bc_network.graphml` (network); `bc_centrality.csv`, `bc_community_corum.csv` (topology and communities); `bc_randomization_test.csv` (degree-preserving null); `bc_enrich_corum.csv`, `bc_enrich_reactome.csv`, `bc_enrich_go.csv`, `bc_enrich_go_specific.csv` (enrichment); `bc_linkpred_metrics.csv`, `bc_missing_components.csv` (link prediction and missing-component control); `bc_drug_classification.csv`, `bc_approved_drugs_per_target.csv` (druggability); `bc_centrality_druggability_corr.csv`, `bc_centrality_druggability_pertarget.csv`, `bc_swisnf_clinical_compounds.csv` (centrality-druggability correlation and verified SWI/SNF clinical

compounds); and `bc_crosscellline_sets.csv`, `bc_crosscellline_summary.json` (cross-cell-line set operations). Figures 1-8 correspond to `bc_fig_network.png`, `bc_fig_centrality.png`, `bc_fig_randomization.png`, `bc_fig_enrichment.png`, `bc_fig_linkpred.png`, `bc_fig_drug_classification.png`, `bc_fig_crosscellline.png`, and `bc_fig_centrality_druggability.png`. Reference resources: CORUM 5.3, Reactome, Gene Ontology, STRING v12 (physical, score ≥ 700), DGIdb, Open Targets, and ChEMBL.

All code required to regenerate every figure and the compiled paper from the raw NDEx network, together with the LaTeX source, figure files, a pinned Python environment specification, and a README with exact run instructions (raw data $\rightarrow$ figures $\rightarrow$ compiled PDF), accompanies this paper as a reproducibility archive intended for public deposit; replace the placeholder URL <https://github.com/USERNAME/REPO> with the deposit location once assigned. The archive is structured so that a third party can reproduce the paper from a clean checkout. Analyses were run in Python 3.13 (networkx, python-igraph, python-louvain, scikit-learn, gensim, goatools, statsmodels, scipy); all random procedures use a fixed seed (42), with the documented exception of multi-threaded node2vec/Word2Vec training (see Methods).

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